

3 Antimicrobial chemicals can be used in food preservation to reduce the risk of food poisoning.

Antimicrobial chemicals derived from plants may be used in food preservation.

An investigation was carried out to assess the antimicrobial properties of extracts from five plants: clove, cumin, ginger, pomegranate and thyme.

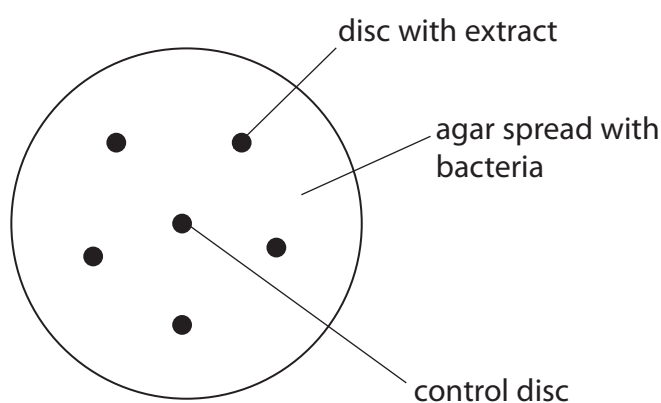
Two types of bacteria were used, type **A** and type **B**.

Filter paper discs were placed in each extract for 10 minutes and then allowed to dry.

Five discs, one for each extract, were placed onto an agar plate spread with bacteria of type **A**. This was repeated for bacteria of type **B**.

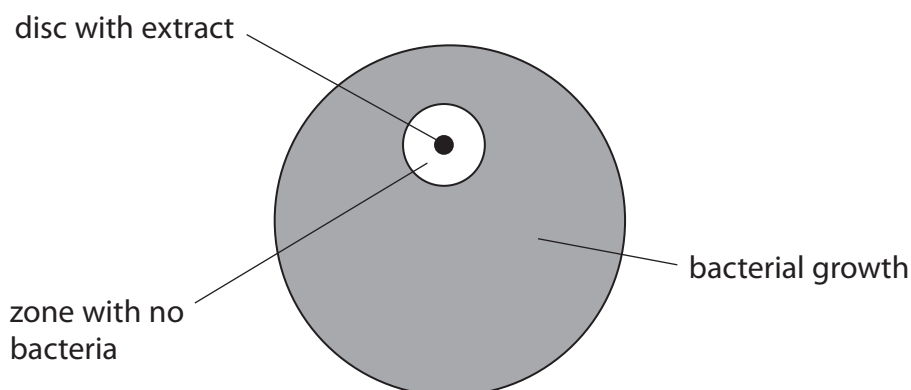
A control disc was added to each plate.

The diagram shows one agar plate after this procedure was completed.



These plates were kept at 5 °C for two hours, and then incubated at 30 °C for 24 hours.

The diagram shows a typical result for one disc.



After this time, the antimicrobial effect was assessed by determining the area of the zone with no bacteria (zone of inhibition).

(a) (i) State the dependent variable in this investigation.

(1)

(ii) Suggest why the plates were kept at 5 °C, before they were incubated.

(1)



(iii) Describe how the area of a zone of inhibition could be determined.

(2)

(iv) Describe the control disc used in this investigation.

(2)

(b) The table shows the results of this investigation.

Plant extract	Area of zone of inhibition / mm ²	
	Bacteria type A	Bacteria type B
clove	196	111
cumin	71	0
ginger	186	0
pomegranate	269	158
thyme	243	0



- (i) Calculate the percentage difference between the inhibitory effect of clove extract on type **A** and type **B** bacteria.

(2)

Answer%

- (ii) Plot a suitable graph to show the results of this investigation.

(5)



(iii) Comment on the results of this investigation into the preservation of food using plant extracts.

(3)

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(c) This method can be used to suggest that an extract has an antimicrobial effect. Other factors can affect the size of the zone of inhibition.

State **two** factors, other than the antimicrobial effect, that could affect the area of the zone of inhibition in this investigation.

(2)

1

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2

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(Total for Question 3 = 18 marks)

TOTAL FOR PAPER = 50 MARKS

